

Employee Attrition Prediction – Project Summary

Introduction

The objective of this project was to understand the factors that influence employee attrition and identify employees who may be at risk of leaving the organization. By analyzing historical employee data, the project aimed to provide meaningful insights that can help the Human Resources (HR) team improve employee retention and make better workforce planning decisions.

Key Findings

The analysis showed that the Research & Development department experienced the highest number of employee exits, followed by the Sales department. Among different job roles, Laboratory Technicians, Sales Executives, and Research Scientists had relatively higher attrition. Employees with lower monthly income, less work experience, and frequent overtime were more likely to leave the organization. Employees who had worked for fewer years at the company also showed a higher tendency to resign compared to long-term employees.

A predictive model was developed to identify employees who may be at risk of leaving. The best-performing model achieved an accuracy of approximately 85%, demonstrating that employee attrition can be predicted with good reliability using available workforce information. The analysis identified Monthly Income, Age, Total Working Years, Overtime, and Years with Current Manager as the most influential factors affecting employee retention.

Business Recommendations

Based on these findings, HR should focus on retaining employees in the Research & Development and Sales departments by improving career development opportunities, monitoring overtime, and conducting regular employee engagement activities. Special attention should also be given to younger employees and employees with shorter tenure through mentoring, training, and clear career progression plans. Reviewing compensation policies and maintaining a healthy work-life balance can further reduce employee turnover and improve overall employee satisfaction.

Conclusion

This project demonstrates that data-driven analysis can help HR identify potential attrition risks before employees decide to leave. While the model should support—not replace—human decision-making, it can serve as an effective tool for planning targeted retention strategies. By acting on these insights, the organization can reduce employee turnover, improve workforce stability, and create a more satisfied and productive work environment.